

ADA Demographic Methods

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Week 9: More Bayes

Where are we at

Bayes rule:

$$p(\theta|y) = \frac{p(y|\theta)p(\theta)}{p(y)}$$

or

$$p(\theta|y) \propto p(y|\theta)p(\theta)$$

- ▶ We are interested in parameter(s) θ
- ▶ Bayesian inference centres on the posterior distribution $p(\theta|y)$
- ▶ This distribution is informed by both prior knowledge (encoded by the prior distribution, $p(\theta)$) and information from data (encoded by the likelihood, $p(y|\theta)$)

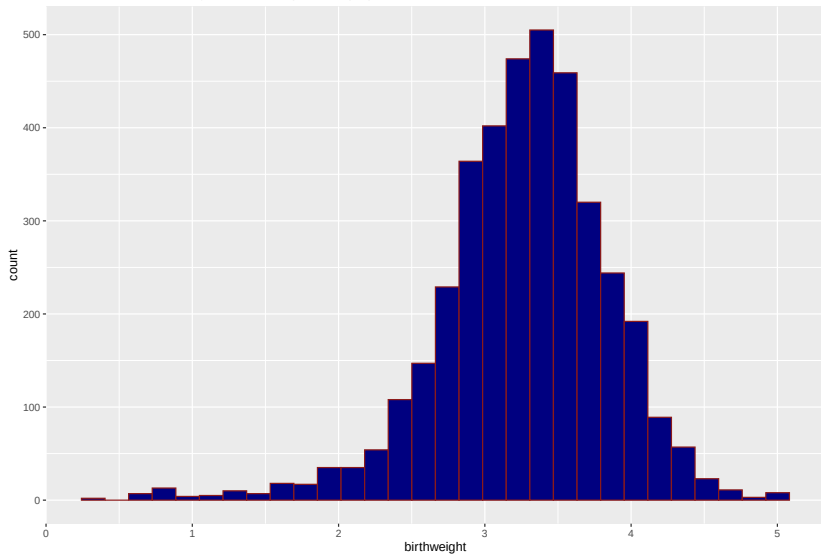
Where are we going

- ▶ Bayesian inference in the regression context
- ▶ How to fit these things (Math is hard, let's sample via MCMC)
- ▶ More on why would we actually do this

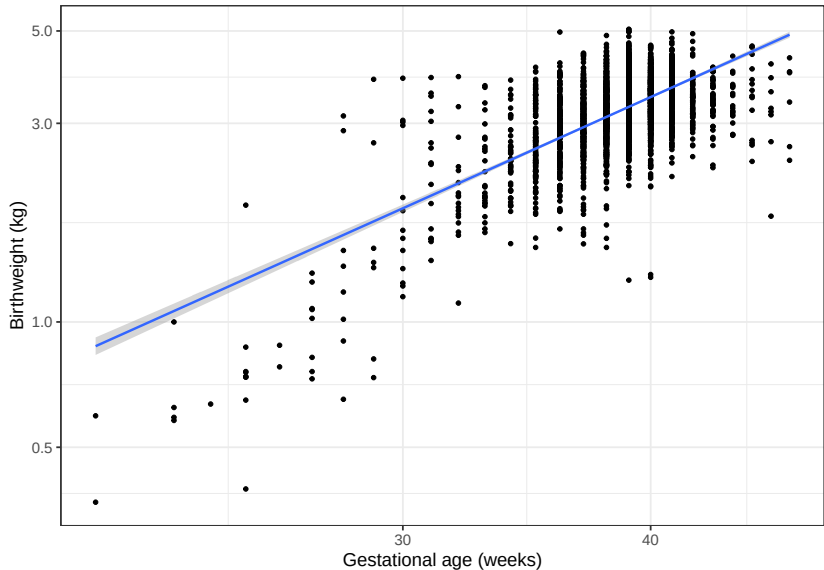
Bayesian regression

Motivating example

Distribution of birth weight, US 2017 (1% sample)



Relationship between birth weight and gestational age



Proposed regression model

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

- ▶ X_i is logged gestational age
- ▶ Y_i is logged birthweight
- ▶ Assume errors $\varepsilon_i \sim N(0, \sigma^2)$

Alternatively we could write this as

$$Y_i | \mu, \sigma^2 \sim N(\mu, \sigma^2)$$

$$\log(\mu) = \beta_0 + \beta_1 X_i$$

Data model

$$Y_i | \mu, \sigma^2 \sim N(\mu, \sigma^2)$$

$$\mu = \beta_0 + \beta_1 X_i$$

- ▶ $Y_i | \mu, \sigma^2 \sim N(\mu, \sigma^2)$ is the data model
- ▶ We are assuming that the Y 's are generated from a distribution that is centered at μ with some variation around the mean governed by σ^2

$$Y_i | \mu, \sigma^2 \sim N(\mu, \sigma^2)$$

$$\mu = \beta_0 + \beta_1 X_i$$

- ▶ The parameters of interest are β_0, β_1 (and to a lesser extent σ)
- ▶ In the Bayesian context, we need to put priors on these in order to estimate them

Distributions

Here, $\theta = \{\beta_0, \beta_1, \sigma\}$

- ▶ Our likelihood $p(y|\beta_0, \beta_1, \sigma)$ is Normal
- ▶ Priors are $p(\beta_0)$, $p(\beta_1)$, $p(\sigma)$
- ▶ Pragmatically, often chosen as Normal, centered at 0 with scale depending on scale of variables
- ▶ If covariates are standardized, a standard Normal ($N(0, 1)$) is seen to be 'weakly informative'
- ▶ For σ , a standard half normal (positive numbers only) is common

Distributions

- ▶ The (joint) posterior is

$$p(\beta_0, \beta_1, \sigma | y) \propto p(y | \beta_0, \beta_1, \sigma) \times p(\beta_0) \times p(\beta_1) \times p(\sigma)$$

- ▶ And we are often interested in the marginal posterior distributions, e.g.

$$p(\beta_1 | y)$$

- ▶ It turns out that even in very simple cases this posterior is hard to write down in closed form
- ▶ So how do we do inference about parameters based on the posterior if we can't write it down?

The solution: MCMC

- ▶ As long as we can get samples from the posterior distribution, we can do inference, using the samples and Monte Carlo approximation
- ▶ How do we sample from a distribution we don't even know?
- ▶ Short answer: clever computational statistics
- ▶ Markov Chain Monte Carlo (MCMC) algorithms are a class of algorithms that allow you to generate samples from the posterior distribution

Monte Carlo approximation

Imagine if I gave you 100 random samples from a Normal distribution.

- ▶ How would you estimate the mean of the distribution?
- ▶ How would you estimate the standard deviation of the distribution?
- ▶ If I instead gave you 1000 samples, could you do a better job at estimating these parameters?

Back to regression

We have

$$Y_i | \mu, \sigma^2 \sim N(\mu, \sigma^2)$$

$$\mu = \beta_0 + \beta_1 X_i$$

with priors

$$\beta_0 \sim N(0, 1), \beta_1 \sim N(0, 1), \sigma \sim N^+(0, 1)$$

Results

Family: gaussian
Links: mu = identity
Formula: log(birthweight) ~ log(gest)
Data: ds (Number of observations: 3842)
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
total post-warmup draws = 4000

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	-6.64	0.15	-6.93	-6.33	1.00	2703	2879
loggest	2.14	0.04	2.05	2.22	1.00	2705	2857

Further Distributional Parameters:

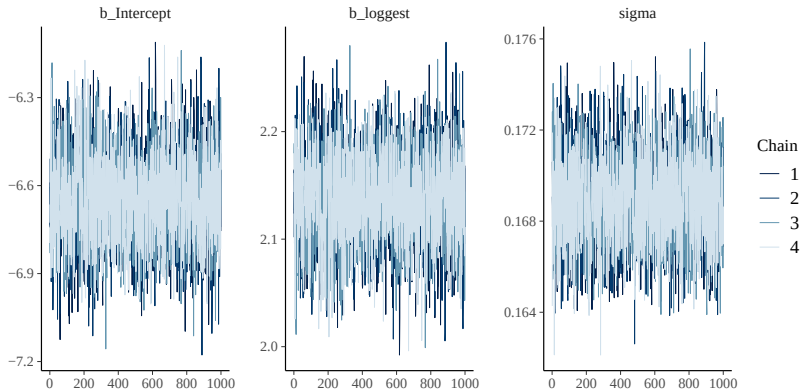
	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sigma	0.17	0.00	0.17	0.17	1.00	2984	2868

Samples

A tibble: 4,000 × 3

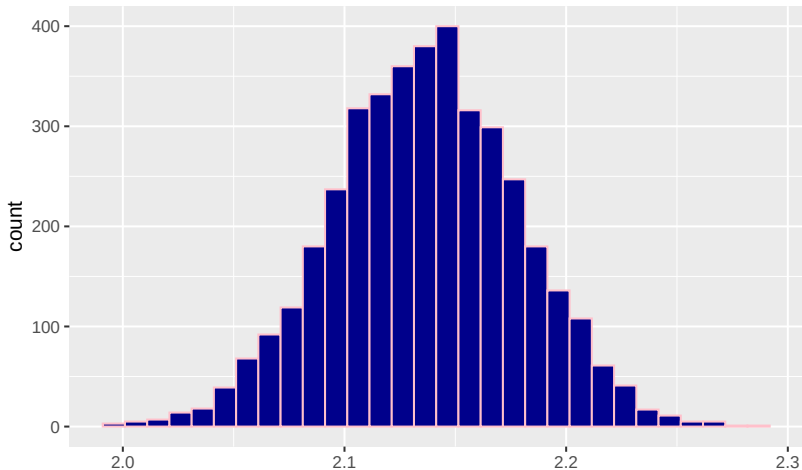
b_Intercept <dbl>	b_loggest <dbl>	sigma <dbl>
-6.742167	2.164819	0.1706081
-6.589697	2.123910	0.1698963
-6.610993	2.128758	0.1686576
-6.698752	2.153685	0.1708516
-6.463061	2.088684	0.1706491
-6.593286	2.124978	0.1680979
-6.638155	2.136933	0.1679719
-6.603619	2.127566	0.1684955
-6.861992	2.197443	0.1658316
-6.689122	2.150824	0.1687911

Samples



Samples

Distribution of samples of coefficient of $\log(\text{gestation})$

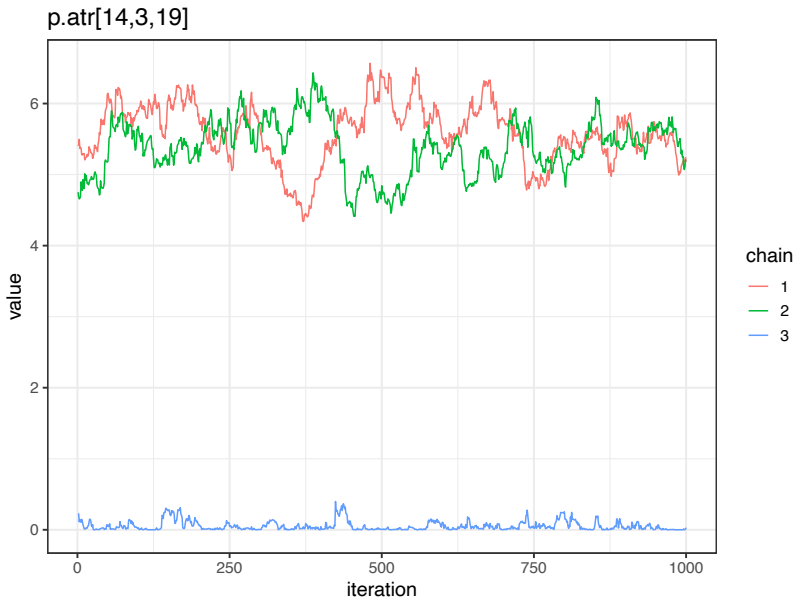


Notes on MCMC

- ▶ In theory, MCMC samplers will always converge to sampling from what we want (the posterior distribution)
- ▶ In practice, this may not be the case
- ▶ What to check: $\hat{R}$ (< 1.05), traceplots (fuzzy caterpillars)

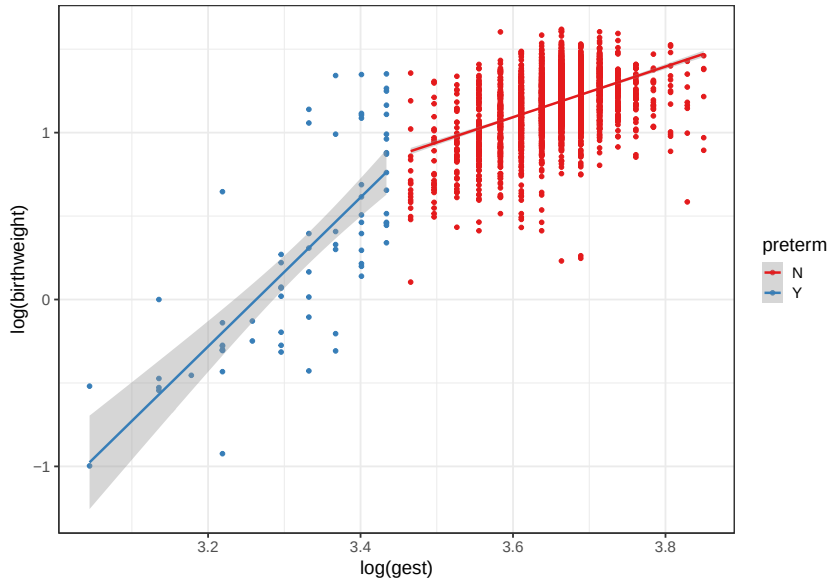
A bad trace plot

Points to model misspecification



Model comparison and selection

Maybe the relationship with age is piecewise linear



Model with interaction

$$Y_i \sim N(\mu, \sigma^2)$$

$$\mu = \beta_0 + \beta_1 \log(X_i) + \gamma_0 Z_i + \gamma_1 \log(X_i) Z_i$$

- ▶ where Z_i is an indicator as to whether or not the baby is preterm.
- ▶ we can fit this in a similar way to the first model: just have two extra parameters (which need priors)

Results

Interpretation?

Family: gaussian
Links: mu = identity
Formula: log(birthweight) ~ log(gest) * preterm
Data: ds (Number of observations: 3842)
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
total post-warmup draws = 4000

Regression Coefficients:

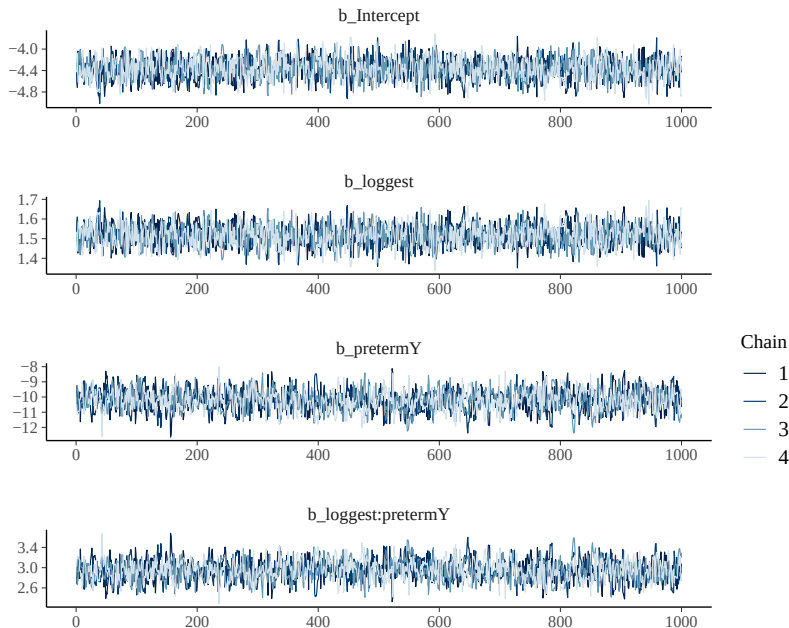
	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	-4.37	0.19	-4.75	-3.99	1.00	1696	2204
loggest	1.52	0.05	1.41	1.62	1.00	1696	2217
pretermY	-10.21	0.67	-11.52	-8.90	1.00	1059	1799
loggest:pretermY	2.95	0.20	2.56	3.34	1.00	1072	1803

Further Distributional Parameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sigma	0.16	0.00	0.16	0.16	1.00	2369	1902

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

Diagnostics of model convergence



Metrics for model selection

- ▶ You may remember AIC/BIC as a way of assessing overall model (out of sample fit)
- ▶ In the Bayesian world, common to use (an approximation of) leave-one-out cross validation (LOO-CV)

LOO-CV intuition:

- ▶ Remove i th data point, rerun the model, then try and predict i th data point
- ▶ Repeat for every single data point
- ▶ Sum up a measure of predictive accuracy (this is called elpd [expected log pointwise predictive density])
- ▶ The model with the higher elpd is better

Comparing the two models

	elpd_diff	se_diff
mod2	0.0	0.0
mod	-175.1	36.3

Two examples where Bayesian regression is
useful

Case 1: dynamic linear regression

Back to Gompertz

Recall that the Gompertz model is the following two-parameter model for mortality:

$$\mu(x) = \alpha e^{\beta x}$$

where $\mu(x)$ is the instantaneous mortality rate (hazard) at age x . This model assumes that mortality increases exponentially with age, which is a pretty good assumption for adult mortality (above age 40 or so, with the exception of older ages).

Write as

$$\log \mu(x) = \log \alpha + \beta x$$

Bayesian Gompertz model

Let's model death counts as a Poisson process where the mean mortality rate has a Gompertz form:

$$y_x \sim \text{Poisson}(P_x \cdot m_x)$$

where y_x is deaths at age x and P_x is population. Model mortality rates m_x as

$$\log m_x = \log \alpha + \beta x$$

We need priors on????

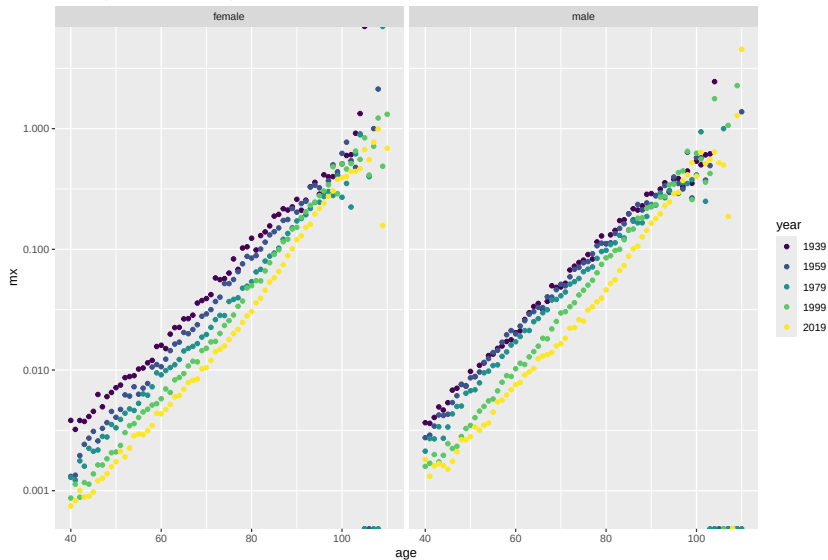
Priors

$$\log \alpha \sim N(0, 10^2)$$

$$\beta \sim N(0, 0.1^2)$$

Let's use some Canadian data

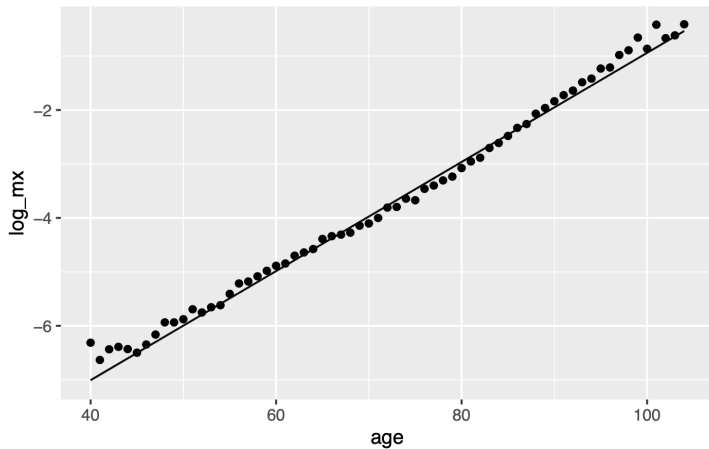
Mortality rate for Ontario by sex, 1939–2019



Model for men in 2019

First let's fit a Gompertz model to males aged 40+ in 2019

Data and fit



Model over time

Let's fit a slightly more complicated model, for multiple years, where the coefficients themselves are modeled as a random walk over time, i.e.

$$\alpha_t \sim N(\alpha_{t-1}, \sigma_\alpha^2)$$

$$\beta_t \sim N(\beta_{t-1}, \sigma_\beta^2)$$

That is, a different set of Gompertz parameters are fit to every year, but we are assuming that the values in the current year are related to those in the previous year. This is a form of dynamic linear regression.

Why is this useful?

So the full model is

$$y_{x,t} \sim \text{Poisson}(P_{x,t} \cdot m_{x,t})$$

$$\log m_{x,t} = \log \alpha_t + \beta_t x$$

$$\alpha_t \sim N(\alpha_{t-1}, \sigma_\alpha^2)$$

$$\beta_t \sim N(\beta_{t-1}, \sigma_\beta^2)$$

With priors

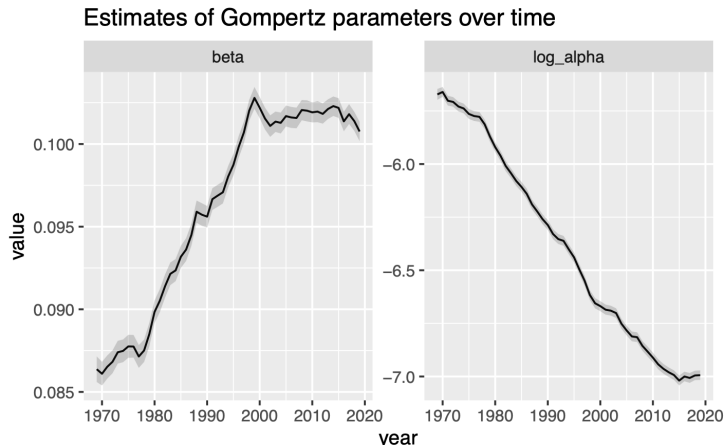
$$\log \alpha_1 \sim N(-6, 1)$$

$$\beta_1 \sim N(0.1, 0.1^2)$$

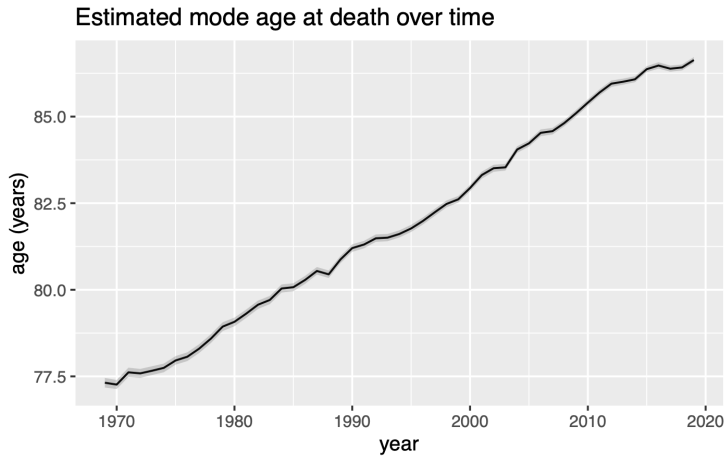
$$\sigma_\alpha, \sigma_\beta \sim N^+(0, 1)$$

Note priors on first time point!

Parameter estimates over time



Modal age of death over time



Case 2: Partial data observed

Mortality across different areas

- ▶ Imagine now we are interested in modeling mortality across (five) geographic areas (e.g. provinces, countries. . .)
- ▶ For four of those areas we have deaths and population by age
- ▶ For the last area we only have a summary of mortality: the probability of dying between ages 40 and 60, or ${}_{20}q_{40}$.
- ▶ Ideally we would like to incorporate this info into the estimates for the fifth area.
- ▶ This happens a lot!!

How would you do this is a classical regression framework?

One proposal

We have a Gompertz model for each area a :

$$\log m_{x,a} = \log \alpha + \beta x$$

We can express the probability of survival in an age group as

$${}_1p_x = e^{-m_x}$$

and then can turn this into a probability of death over the interval 40-60 using:

$${}_{20}q_{40} = 1 - \prod_{x=40}^{60} {}_1p_x$$

Two data models

Now we have two data models. For the four regions with complete data, we have

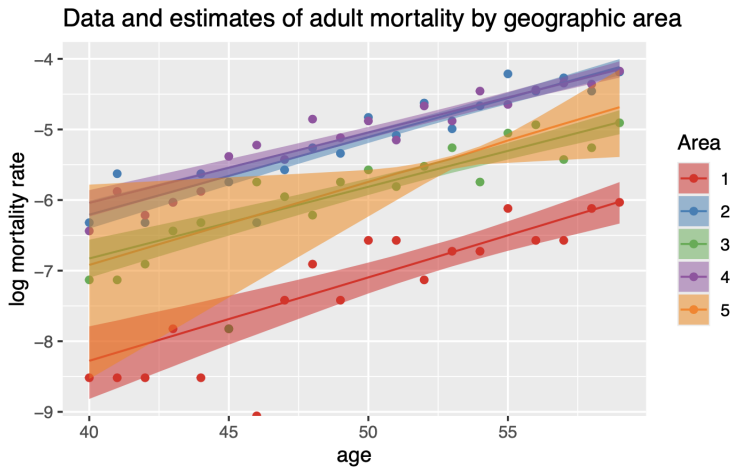
$$y_{x,a} \sim \text{Poisson}(P_{x,a} \cdot m_{x,a})$$

For the fifth region with a summary, put these data in as a Normal:

$$z_5 \sim N({}_{20}q_{40}(5), 0.001^2)$$

- ▶ where z_5 is the observed value of ${}_{20}q_{40}$ for the fifth region
- ▶ Note that I put very small variance around this observed value; in practice you may have info from sampling error, etc

Results



Example from the wild

RESEARCH ARTICLE | APRIL 01 2022

Competing Effects on the Average Age of Infant Death

Monica Alexander; Leslie Root

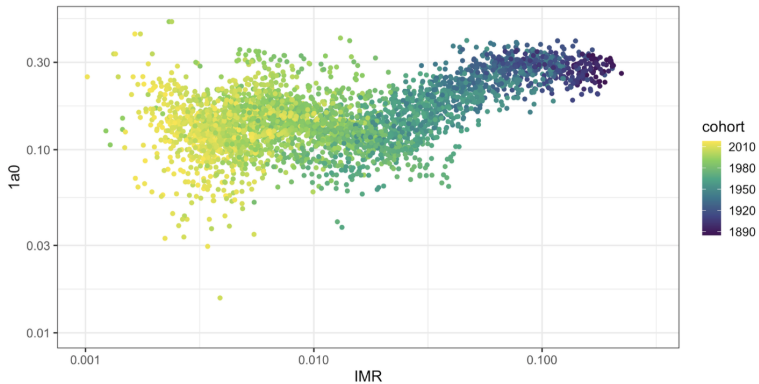
Demography (2022) 59 (2): 587–605.

Motivation

- ▶ We need values of ${}_1a_0$ for the life table
- ▶ This is average age of infant death
- ▶ Usually have to approximate because we don't have data
- ▶ Existing approximations look at relationship between average age and overall infant mortality

Average age of infant death versus infant mortality rate

Cohorts born in 1885-2018

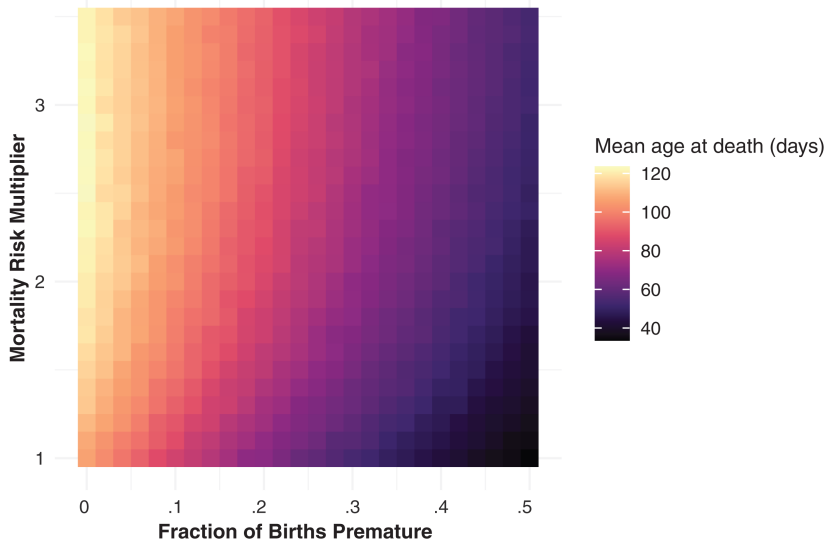


Source: HMD

It's linear until it's not

- ▶ Existing approximations add a 'cutpoint' and fit a piecewise linear regression model
- ▶ We argued that as mortality conditions improve, the relationship is muddled because of prematurity

c. Competing effects on average age of infant death



A ratio as a proxy

- ▶ Data on prematurity is not readily available
- ▶ But data on infant mortality and under-five mortality is
- ▶ Use this ratio (IMR/U5MR) as a proxy for prematurity



Compare with existing best model

$${}_1a_{0i} = \alpha + \beta_1(IMR_i) + \beta_2\left(\frac{IMR_i}{U5MR_i}\right) + \varepsilon_i.$$

(1)

For reference, existing models for ${}_1a_0$ (as in [Andreev and Kingkade \(2015\)](#))⁵ have the form of a piecewise linear relationship with ${}_1q_0$ with one “cutpoint” (θ) at which the slope of the linear relation with ${}_1q_0$ changes:⁶

$${}_1a_{0i} = \alpha + \beta_1({}_1q_{0i}) + \beta_2({}_1q_{0i} - \theta) + \varepsilon_i.$$

(2)

- ▶ Look at in- and out-of-sample error
- ▶ Utilize Bayesian framework for model evaluation metrics and uncertainty propagation

Results

Table 4 In- and out-of-sample root-mean-square errors for different models

Type	Sex	AK	Piecewise	Ratio
In-sample	F	0.038	0.038	0.037
In-sample (AK data)	F	0.035	0.038	0.036
Out-of-sample	F	0.038	0.038	0.036
In-sample	M	0.036	0.036	0.036
In-sample (AK data)	M	0.042	0.038	0.038
Out-of-sample	M	0.044	0.041	0.040
In-sample	Both	—	0.038	0.037
Out-of-sample	Both	—	0.041	0.038

Note: AK = Andreev and Kingkade.

Table 5 Root-mean-square errors (RMSEs) of piecewise and ratio models fitted to U.S. data by sex and race/ethnicity

Race	Sex	Model	RMSE
NHB	F	Ratio	0.0090
NHB	F	Piecewise	0.0409
NHW	F	Ratio	0.0129
NHW	F	Piecewise	0.0689
NHB	M	Ratio	0.0080
NHB	M	Piecewise	0.0355
NHW	M	Ratio	0.0139
NHW	M	Piecewise	0.0710

Note: NHB = non-Hispanic Black. NHW = non-Hispanic White.